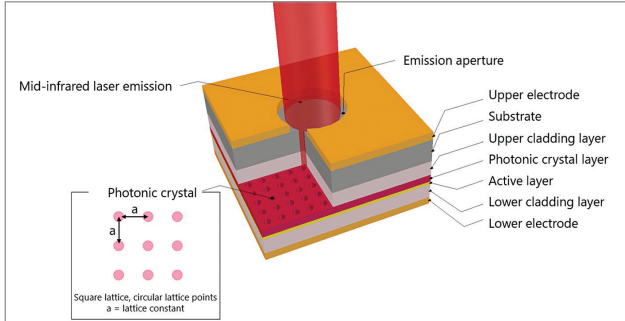


Smaller, more efficient gas sensors

A newly developed infrared laser technology by Asahi Kasei Microdevices Corporation (AKM), in collaboration with the Kyoto University Institute for Advanced Study, could enable smaller, more efficient gas sensors by integrating

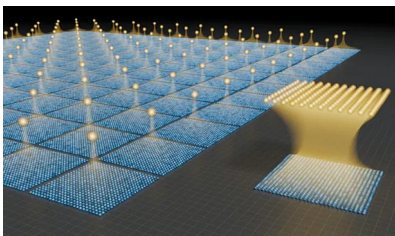


Breakthrough in PCSEL Optical Sensing Technology by Asahi Kasei Microdevices enables 2µm-band Infrared Laser Generation (Credit: <https://www.akm.com>)

high-performance light sources into compact devices. The innovation uses advanced photonic designs to deliver precise, narrow-linewidth infrared emission essential for accurate gas detection. Engineered through research conducted in Japan, the system leverages photonic crystal surface-emitting lasers to achieve high brightness and functionality in a miniaturised form factor. The advancement could support portable environmental monitoring, industrial safety, and medical diagnostics.

Quantum computers may need far fewer qubits

Quantum computers may require far fewer qubits than previously thought, thanks to advances in algorithms and error-correction techniques that significantly reduce hard-



Shor's algorithm can break modern cryptographic schemes (Image Credit: Caltech/Robert Hurt, IPAC-SELab)

ware demands. Researchers show that optimised designs can achieve the same computational goals with dramatically smaller qubit counts. Developed through studies conducted by scientists at the California Institute of Technology (Caltech), Pasadena, California, USA, the approach focuses on improving quantum circuit efficiency and fault tolerance rather than simply scaling hardware. The finding could accelerate practical quantum computing by lowering cost and complexity, making systems easier to build and deploy. It builds on a broader shift in the field, where performance increasingly depends on qubit quality and error correction instead of sheer quantity.

AI system tracks sleep without cameras

An AI-powered system can track sleep patterns and posture without using cameras or wearable devices, offering a privacy-preserving alternative for monitoring. The technology relies on non-invasive sensing and machine learning to analyse breathing, movement, and body position in real time. Developed by a team at the National Institute of Technology Rourkela in India, the system uses advanced sensors embedded in the environment to capture sleep data accurately without direct contact. The innovation could improve the diagnosis of sleep disorders and enable continuous home monitoring. It builds on efforts to replace intrusive camera- and wearable-based systems with more comfortable, contactless solutions.

PCSEL enables real-world optical communication

Photonic crystal surface-emitting lasers (PCSELs) have been demonstrated in real-world optical communication for the first time, marking a step toward high-speed, secure



Real world optical communication (Credit: <https://www.record-courier.com>)

wireless data links. The breakthrough shows these semiconductor lasers can operate outside laboratory conditions while maintaining stable, long-distance transmission. Developed by Vector Photonics in collaboration with Fraunhofer

UK Research Ltd, the system uses PCSELs to emit narrow, high-brightness, low-divergence beams, reducing signal loss and improving link reliability. The advance could enable faster free-space optical networks for telecom and defence, building on efforts to replace conventional radio-frequency communication with high-capacity optical links.

Real-time traffic emission measurements made possible

Researchers at Massachusetts Institute of Technology (MIT), USA, have developed a real-time framework to measure traffic emissions at the block level. By combining traffic camera feeds, mobile phone data, and emission models, the system maps changes in pollution across city streets with high precision. The study shows that policy impacts, including congestion pricing, can be quantified quickly, offering urban planners a powerful tool for climate and transportation decisions and making cities smarter planning systems.